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Carefully read the question and design the **BankAccount** class such that the following main method produces the expected output.

Tester Code	Expected Output
<pre> public class BankAccountTester { public static void main(String[] args) { BankAccount b1 = new BankAccount("Rafi", 1000); BankAccount b2 = new BankAccount("Nina", 300); System.out.println("1=====1"); System.out.println(b1.details()); System.out.println(b2.details()); System.out.println("2=====2"); b1.deposit(500); b2.deposit(-100); System.out.println(b1.details()); System.out.println(b2.details()); System.out.println("3=====3"); b1.withdraw(200); b2.withdraw(500); System.out.println(b1.details()); System.out.println(b2.details()); System.out.println("4=====4"); b1.transfer(b2, 300); System.out.println(b1.details()); System.out.println(b2.details()); System.out.println("5=====5"); BankAccount b3 = new BankAccount(b2); b3.setName("Nina Clone"); b3.deposit(200); System.out.println(b2.details()); System.out.println(b3.details()); System.out.println("6=====6"); BankAccount b4 = b1; b4.deposit(100); System.out.println(b1.details()); System.out.println(b4.details()); } } </pre>	<pre> Account opened for Rafi. Account opened for Nina. 1=====1 Rafi has 1000 taka. Nina has 300 taka. 2=====2 Invalid deposit! Rafi has 1500 taka. Nina has 300 taka. 3=====3 Insufficient balance! Rafi has 1300 taka. Nina has 300 taka. 4=====4 Rafi transferred 300 taka to Nina. Rafi has 1000 taka. Nina has 600 taka. 5=====5 Account opened for Nina. Nina has 600 taka. Nina Clone has 800 taka. 6=====6 Rafi has 1100 taka. Rafi has 1100 taka. </pre>

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Carefully read the question and design the **MobilePhone** class such that the following main method produces the expected output.

Tester Code	Expected Output
<pre> public class MobileTester { public static void main(String[] args) { MobilePhone m1 = new MobilePhone("Samsung", 60); MobilePhone m2 = new MobilePhone("iPhone", 90); System.out.println("1=====1"); System.out.println(m1.details()); System.out.println(m2.details()); System.out.println("2=====2"); m1.charge(30); m2.charge(-10); System.out.println(m1.details()); System.out.println(m2.details()); System.out.println("3=====3"); m1.useBattery(50); m2.useBattery(120); System.out.println(m1.details()); System.out.println(m2.details()); System.out.println("4=====4"); m1.shareBattery(m2, 20); System.out.println(m1.details()); System.out.println(m2.details()); System.out.println("5=====5"); MobilePhone m3 = new MobilePhone(m1); m3.setBrand("Samsung Copy"); m3.charge(10); System.out.println(m1.details()); System.out.println(m3.details()); System.out.println("6=====6"); MobilePhone m4 = m2; m4.charge(5); System.out.println(m2.details()); System.out.println(m4.details()); } } </pre>	<pre> Phone Samsung created. Phone iPhone created. 1=====1 Samsung battery: 60% iPhone battery: 90% 2=====2 Invalid charge amount! Samsung battery: 90% iPhone battery: 90% 3=====3 Battery too low! Samsung battery: 40% iPhone battery: 90% 4=====4 Samsung shared 20% battery with iPhone. Samsung battery: 20% iPhone battery: 110% 5=====5 Phone Samsung created. Samsung battery: 20% Samsung Copy battery: 30% 6=====6 iPhone battery: 115% iPhone battery: 115% </pre>

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Carefully read the question and design the Membership class with appropriate variables and methods such that the following main method produces the expected output. For simplicity, assume that only Regular and Premium membership types are supported here. Note that, when not given a specific discount rate, a Regular membership receives a 20% discount and a Premium membership receives a 40% discount.

Tester Code	Expected Output
<pre> public class MembershipTester { public static void main(String[] args) { System.out.println(Membership.status()); System.out.println("1 >====="); Membership m1 = new Membership("Rahim", "Regular", 2500); Membership m2 = new Membership("Karim", "Premium", 5000); System.out.println(m2.details()); System.out.println(m1.details()); System.out.println("2 >====="); System.out.println(Membership.status()); System.out.println("3 >====="); m1.cancelMembership(); m2.discount(0.80); Membership m3 = new Membership("Nadia", "Regular", 3000); System.out.println(Membership.status()); System.out.println("4 >====="); System.out.println(m3.details()); System.out.println(m2.details()); System.out.println(m1.details()); System.out.println("5 >====="); m3.discount(); m2.discount(); Membership m4 = new Membership("Sakib", "Premium", 6000); System.out.println("6 >====="); System.out.println(m4.details()); System.out.println(Membership.status()); } } </pre>	<pre> Total memberships: 0 Regular and 0 Premium Memberships Total fee: 0.0 Tk 1 >===== Membership ID: P-002 Premium: Karim => 5000.0 Tk Membership ID: R-001 Regular: Rahim => 2500.0 Tk 2 >===== Total memberships: 1 Regular and 1 Premium Memberships Total fee: 7500.0 Tk 3 >===== R-001 membership has been cancelled. P-002 fee reduced to 4000.0 Tk Total memberships: 2 Regular and 1 Premium Memberships Total fee: 7000.0 Tk 4 >===== Membership ID: R-003 Regular: Nadia => 3000.0 Tk Membership ID: P-002 Premium: Karim => 4000.0 Tk Membership ID: R-001 Regular: Cancelled => 0.0 Tk 5 >===== R-003 fees reduced to 2400.0 Tk P-002 fees reduced to 2400.0 Tk 6 >===== Membership ID: P-004 Premium: Sakib => 6000.0 Tk Total memberships: 2 Regular and 2 Premium Memberships Total fee: 10800.0 Tk </pre>

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1	public class Gandalf {
2	public void method1(){
3	System.out.println("Gandalf 1");
4	}
6	public void method2(){
7	System.out.println("Gandalf 2");
8	method1();
9	}
10	}
11	public class Bilbo extends Gandalf{
12	public void method1(){
13	System.out.println("Bilbo 1");
14	}
15	}
16	public class Gollum extends Gandalf{
17	public void method3(){
18	System.out.println("Gollum 3");
19	}
20	}
21	public class Frodo extends Bilbo{
22	public void method1(){
23	System.out.println("Frodo 1");
24	super.method1();
25	}
26	
27	public void method3(){
28	System.out.println("Frodo 3");
29	}
30	}

Assuming the following variables have been defined:

```
Gandalf var1 = new Frodo();
Gandalf var2 = new Bilbo();
Gandalf var3 = new Gollum();
Object var4 = new Frodo();
Bilbo var5 = new Frodo();
Object var6 = new Gollum();
```

Use a/b/c for multi-line output

	Statement	Output
1	var5.method2();	
2	var6.method2();	
3	((Frodo)var4).method3();	
4	((Frodo)var6).method2();	
5	((Gollum)var1).method3();	
6	((Gollum)var4).method1();	
7	((Gandalf)var1).method2();	
8	((Frodo)var4).method1();	
9	((Gollum)var6).method2();	
10	((Gandalf)var2).method1();	
11	((Bilbo)var6).method2();	
12	((Frodo)var1).method3();	
13	((Gandalf)var5).method3();	

```

1 public class A {
2     public static int temp = 2;
3     public int x = 3, y = 7, sum = 0;
4     public A() {
5         y -= temp++;
6         sum = x + y - temp;
7     }
8     public A(int x) {
9         y += x;
10        temp--;
11    }
12    public String methodA(int m, int n) {
13        int z = x + m + (++temp);
14        y = y + n - z;
15        sum = sum + x + y;
16        return y + " " + x + " " + sum;
17    }
18 }
19 public class B extends A {
20     public static int temp = 8;
21     public int x = 3, y = 5;
22     public B() {
23         super(2);
24         y = y - temp;
25         sum = sum - x + y;
26     }
27     public String methodA(int m, int n) {
28         int z = 0;
29         z = x + m + (temp++);
30         y = y - n + z;
31         System.out.println(super.methodA(x, y));
32         sum = sum + x + y + temp;
33         return x + " " + y + " " + sum;
34     }
35     public void methodB(int m) {
36         int y = m + this.y;
37         x = x + y + temp;
38         System.out.println(methodA(y, x));
39         sum = sum + x + y;
40         super.methodA(sum, m);
41         System.out.println(x + " " + y + " " + sum);
42     }
43 }

```

Illustrate the output of the following statements written in the main method of the driver code. [Answers without workings on the script will be rejected]

Driver code:

```

public class Tester{
    public static void
    main(String []args){
        B b1 = new B();
        b1.methodB(4);
    }
}

```

Output:
